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Recursive VAR Identification of Risk-Induced Exchange Rate Dynamics and Evidence from the Yen Carry Trade

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1 Introduction

Persistent deviations from Uncovered Interest Rate Parity (UIRP) represent one of the most enduring puzzles in empirical international finance. According to standard macroeconomic theory, high-yield currencies should depreciate to offset the interest rate differential, thereby equalizing expected returns across borders. In practice, however, high-yield currencies often appreciate or fail to depreciate sufficiently, allowing international investors to generate persistent excess returns through carry trade strategies. To explain this forward premium puzzle, modern financial literature increasingly focuses on time-varying currency risk premiums, market liquidity constraints, and global risk transmission channels.

The real-world relevance of these transmission channels was vividly illustrated on March 31, 2026, when escalating geopolitical tensions in the Middle East (specifically, the outbreak of the Iran war) ignited a severe sell-off across global financial markets (Kihara & Yamaguchi, 2026). The conflict threatened the potential closure of the Strait of Hormuz, a crucial maritime transit chokepoint handling approximately one-fifth of global oil and gas flows, which instantly drove up crude oil prices and triggered a massive flight-to-safety toward the US Dollar. For Japan, this external shock resulted in a highly destabilizing "triple sell-off" in domestic assets, where the Japanese Yen plunged to near the critical 160-per-dollar mark, the Nikkei 225 stock index lost over 11% of its value in March, and panic-driven liquidations in the debt market pushed the benchmark 10-year Japanese Government Bond (JGB) yield to levels not seen since 1999 (Kihara & Yamaguchi, 2026).

Japanese policymakers faced an acute dilemma. Finance Minister Satsuki Katayama branded the yen's sharp decline as "speculative" for the first time since the war began, indicating Tokyo's readiness to intervene "on all fronts" (Kihara & Yamaguchi, 2026). Concurrently, soaring energy import costs and imported inflation from the weak yen piled immense pressure on the Bank of Japan (BOJ) to raise interest rates, despite a fragile domestic economy.

This crisis underscores the central research question. What is the causal effect of exogenous shocks to financial risk indicators (VIX, oil price, and FX risk reversals) on the JPY/USD exchange rate, operating through yen-funded carry trade unwinding and safe-haven demand channels, after controlling for interest rate differentials, FX liquidity, and relative equity performance?

This proposal addresses this question by exploring and testing the following causal relationship.

A one-standard-deviation positive shock to the VIX (equity market fear) or to the 1-month FX risk reversal (hedging cost) causes a statistically significant appreciation of the Japanese Yen against the US Dollar within 2–3 days, with effects persisting over a 7-day horizon. This causal effect operates through two indirect financial risk transmission channels, including (1) unwinding of yen-funded carry trades, and (2) safe-haven portfolio rebalancing (Uncovered Equity Parity). The effect holds after controlling for changes in interest rate differentials, oil prices, bid-ask spreads, and relative equity returns.

Under normal conditions, the yen acts as a primary funding currency for carry trades due to Japan’s historically low interest rates. However, during periods of acute global uncertainty (such as the March 2026 geopolitical shock), global investors rapidly unwind their leveraged positions, purchasing yen to pay off their liabilities. This paper proposes to empirically analyze the dynamics of this indirect financial risk transmission mechanism during the 2025–2026 crisis.

1.1 Research Objectives

- To identify the causal effect of global risk sentiment shocks (measured by VIX) and currency crash risk shocks (measured by FX risk reversals) on the JPY/USD daily exchange rate return over a 7-day horizon.

- To evaluate the relative importance of the carry trade unwinding channel versus the equity portfolio rebalancing channel (Uncovered Equity Parity) in driving short-term yen adjustments.
- To examine whether standard interest rate differentials retain any direct predictive power over JPY/USD dynamics during periods of severe geopolitical stress.

2 Related Literature

To investigate these channels, we build directly upon the empirical and theoretical framework established by Guyot et al. (2026). Their study examines the JPY/USD market using daily data from January 1, 2018, to December 31, 2024. The authors employ a reduced-form vector autoregression (VAR) model with a Cholesky decomposition to capture the endogenous interactions between interest rate differentials, foreign exchange liquidity, global risk indicators (including the VIX, oil price shocks, and currency risk reversals), and relative equity returns (Guyot et al., 2026).

The primary empirical contribution of Guyot et al. (2026) is the confirmation of persistent UIRP deviations and the identification of indirect financial risk transmission channels that explain these deviations. Using Granger causality tests and impulse response functions, the authors reveal several key insights.

1. Interest rate differentials do not directly explain short-term exchange rate movements, confirming the failure of UIRP.
2. Shocks to global risk sentiment (VIX) and options-implied JPY crash risk (risk reversals) exert a highly significant and immediate effect on JPY/USD returns.
3. The relative equity market performance gap (EGAP) serves as a critical transmission intermediary, linking global risk conditions to portfolio-based exchange rate adjustments, consistent with the Uncovered Equity Parity (UEP) hypothesis.

4. The Japanese Yen maintains its role as a key safe-haven currency, experiencing significant appreciation during global uncertainty spikes due to carry trade unwinding dynamics.

The theoretical foundation for these findings rests on three pillars of the international finance literature. First, the carry trade crash literature, beginning with Brunnermeier et al. (2009), establishes that carry trade strategies are subject to crash risk: during periods of sharply rising volatility, leveraged investors are forced to unwind their positions, generating abrupt reversals in funding currencies. Brunnermeier et al. (2009) identify a VIX threshold below which the yen's flight-to-safety effect is strong but weakens under extreme crisis conditions, a nonlinearity that underscores the importance of our subsample analysis. Their work also documents that skewness in currency options (risk reversals) predicts carry trade returns, directly motivating our inclusion of FX risk reversals as a key regressor (Brunnermeier et al., 2009).

Second, the yen-specific carry trade evidence is extensively documented by Gagnon and Chaboud (2007), who analyze episodes of yen-funded carry trade unwinding in October 1998, May 2006, and February 2007. They demonstrate that the yen's exchange rate is acutely sensitive to shifts in global risk sentiment and FX liquidity conditions, and that liquidity shortages during unwinding episodes amplify currency volatility. Their finding that bid-ask spreads widen precisely when carry trades reverse motivates our inclusion of the FX liquidity (spread) channel (Gagnon & Chaboud, 2007).

Third, the safe-haven currency literature, surveyed by Habib and Stracca (2012) and Lee (2017), empirically identifies which currencies serve as stores of value during global crises. Habib and Stracca (2012) find that safe-haven status is associated with low interest rates, large net foreign asset positions, and deep financial markets — all characteristics of the Japanese Yen. Lee (2017) further documents that the yen's safe-haven property is most pronounced during moderate stress periods but may attenuate during tail-risk events, a nonlinearity our GPR-subsampled design is explicitly constructed to test (Habib & Stracca, 2012; Lee, 2017).

While Guyot et al. (2026) construct a robust baseline, their sample terminates in December 2024, prior to the severe Middle East crisis of March 2026. This proposal extends their

framework to the 2025–2026 period to capture how these channels behave under extreme geopolitical stress. Furthermore, due to the lack of access to a Bloomberg Terminal, we modify their data acquisition strategy by utilizing public APIs (FRED and Yahoo Finance) and open academic datasets (Iacoviello’s GPR index) to construct equivalent variables, verifying that public data can replicate and extend their findings.

3 Methodology

This section describes the data sources, variable construction, econometric framework, identification strategy, and implementation details that together enable a replicable analysis of how financial risk transmission channels affect the JPY/USD exchange rate. We structure the exposition to satisfy the standard that a graduate student with general econometric training could reproduce the full analysis after reading this section and accessing the public data sources described herein (Rossi, 2013).

3.1 Data Sources and Provenance

We construct a daily time-series dataset spanning January 1, 2018, through May 31, 2026, yielding approximately 2,100 trading days after alignment across markets. This extends the Guyot et al. (2026) sample by 17 months to capture the 2025–2026 Iran conflict period. Table 1 details all variables, their source series, retrieval methods, and timestamps.

Unlike the original study which sourced all data from the Bloomberg Terminal (a proprietary financial platform requiring a paid subscription), we retrieve every series from freely accessible public databases and APIs (Guyot et al., 2026). This substitution is motivated by two considerations: first, to demonstrate that the core empirical findings are not artifacts of proprietary data; second, to ensure complete reproducibility by any researcher with internet access. All data were retrieved on June 1, 2026.

Table 1: Data Sources, Variable Construction, and Retrieval Protocols

Variable	Proxy / Series Code	Source	Retrieval Method
$\Delta JPY/USD$	DEXJPUS (FRED), USDJPY=X (Yahoo spot)	FRED, Yahoo	FRED API, Yahoo API
$\Delta(i - i^*)$	Japan 1Y JGB (MOF), DFF (US Fed Funds)	MOF Japan, FRED	MOF website, FRED API ^a
ΔY_{Curve}^{Diff}	MOF JGB 10Y/2Y; DGS10, DGS2 (US 10Y/2Y)	MOF Japan, FRED	MOF website, FRED API
$\Delta Brent$	DCOILWTICO (WTI), BZ=F (Brent)	FRED, Yahoo	FRED API, Yahoo API
VIX	CBOE Volatility Index ($\sim VIX$)	Yahoo Finance	Yahoo API
$Spread$	JPY/USD Bid-Ask (OANDA); proxy $(H - L)/C$ (USDJPY=X)	OANDA, Yahoo	OANDA API, Yahoo API ^a
$RiskReversal$	1-Month JPY/USD 25- Δ Risk Reversal	BIS (quarterly)	BIS website ^b
$EGAP$	Nikkei 225 ($\sim N225$), S&P 500 ($\sim GSPC$)	Yahoo Finance	Yahoo API
GPR	Geopolitical Risk Index (daily)	Iacoviello (2026)	Iacoviello website

All series retrieved 2026-06-01, spanning 2018-01-01 through 2026-05-31 (GPR through 2026-05-26). FRED accessed via FRED API (`pandas_datareader` CSV export endpoint) (Federal Reserve Bank of St. Louis, 2026); Yahoo Finance via Yahoo API (`yfinance`) (Yahoo Finance, 2026); MOF Japan JGB yields: https://www.mof.go.jp/english/policy/jgbs/reference/interest_rate/; BIS: (Bank for International Settlements, 2026); Iacoviello GPR: <https://www.matteoiacoviello.com/gpr.htm> (Iacoviello, 2026).

^a Data Substitution: The original Guyot (2026) study used the Japan Call Rate (BOJ uncollateralized overnight) and US Fed Funds Rate from Bloomberg. Daily Japanese short-term bank rates are unavailable from public APIs (FRED publishes them only at monthly frequency); we substitute the daily 1-Year JGB yield from MOF Japan. For the US, we use the Fed Funds Rate (DFF, available daily from FRED), which matches the instrument used by Guyot. For the spread, pending OANDA API access, we employ the intraday range proxy $(High - Low)/Close$ from USDJPY=X, a standard liquidity proxy in market microstructure literature (Corwin & Schultz, 2012; Goyenko et al., 2009).

^b Data Limitation: The 1-month JPY/USD 25-delta risk reversal is published only at quarterly frequency by the BIS. We interpolate to daily via cubic spline and flag this as a limitation relative to the original study's Bloomberg-sourced daily data.

Because Japanese and US financial markets observe different national holidays, we align the dataset by retaining only dates on which all variables have valid observations (Guyot et al., 2026). Specifically, the foreign exchange market for JPY/USD trades 24 hours per day on virtually all business days globally; we define a “valid trading day” as a date on which both the NYSE and TSE are open, following the synchronization procedure in Guyot et al. (2026, p. 11). Dates where any variable is missing are excluded, resulting in a final balanced panel of approximately 1,800 to 2,000 daily observations depending on the sample window.

3.2 Data Preparation and Variable Construction

Each raw series undergoes stationarity pre-tests and transformation before entering the VAR. We apply the Augmented Dickey-Fuller (ADF) test to every series at the 5% significance level, with lag length selected by the Schwarz Criterion ($\max p = 12$). Variables found to contain a unit root are transformed into log-differences or first-differences to achieve $I(0)$ stationarity.

3.2.1 Exchange Rate Return ($\Delta JPY/USD_t$)

Let s_t be the JPY/USD noon buying rate (FRED series DEXJPUS). We compute the continuously compounded daily return as

$$\Delta JPY/USD_t = 100 \times [\ln(s_t) - \ln(s_{t-1})] \quad (1)$$

following the convention in Guyot et al. (2026). A negative value implies yen appreciation (fewer yen per dollar).

3.2.2 Interest Rate Differential ($\Delta(i_t - i_t^*)$)

Let i_t^{JPN} be the yield on the 1-Year Japanese Government Bond (MOF Japan daily fixing) and i_t^{USA} be the US Federal Funds Rate (FRED series DFF). The original study by Guyot et al. (2026) employed Japan’s uncollateralized overnight call rate and the US Fed Funds Rate, both sourced from the Bloomberg Terminal. Because daily Japanese short-term bank rates are not

available from public APIs (FRED publishes Japanese short-term interest rates only at monthly frequency), we substitute the daily 1-Year JGB yield from the Japan Ministry of Finance as a short-term rate proxy. This substitution is consistent with the literature’s finding that short-term interest rate differentials do not significantly explain short-run exchange rate movements (Engel, 2016; Guyot et al., 2026). For the US, we use the Fed Funds Rate (DFF) which is available daily from FRED and corresponds to the precise instrument employed by Guyot. We compute

$$\Delta(i_t - i_t^*) = (i_t^{JPN} - i_t^{USA}) - (i_{t-1}^{JPN} - i_{t-1}^{USA}) \quad (2)$$

as the first-difference of the interest rate differential (Masujima & Sato, 2024). This transformation ensures stationarity and captures the “surprise” component of monetary policy relevant for daily exchange rate dynamics.

3.2.3 Yield Curve Differential ($\Delta YCurve_t^{Diff}$)

Let $YC_t^{JPN} = y_{10Y,t}^{JGB} - y_{2Y,t}^{JGB}$ be Japan’s yield curve slope (10-year minus 2-year JGB yields, sourced from MOF Japan daily fixing data) and $YC_t^{US} = y_{10Y,t}^{TNOTE} - y_{2Y,t}^{TNOTE}$ be the US equivalent (FRED series DGS10 and DGS2). Note that FRED publishes Japanese bond yields (IRLTLT01JPM156N) only at monthly frequency; we therefore use the Ministry of Finance Japan’s daily JGB yield data, which publishes the full yield curve (1Y–40Y) for every Tokyo business day. The relative term premium is

$$\Delta YCurve_t^{Diff} = [YC_t^{JPN} - YC_t^{US}] - [YC_{t-1}^{JPN} - YC_{t-1}^{US}] \quad (3)$$

which captures shifts in the relative risk term structure between Japanese and US bond markets. A positive value indicates that the Japanese bond risk premium is accelerating relative to US bonds, signaling monetary policy divergence (Guyot et al., 2026).

3.2.4 Oil Price Change ($\Delta Brent_t$)

Let $Brent_t$ be the daily closing price of WTI crude oil (DCOILWTICO) or Brent crude (BZ=F) in USD per barrel. We compute

$$\Delta Brent_t = 100 \times [\ln(Brent_t) - \ln(Brent_{t-1})] \quad (4)$$

to capture the daily percentage change in energy prices. While the original study uses Brent exclusively, we adopt WTI as the primary series (Brent for robustness) due to superior FRED data availability. The two benchmarks exhibit a correlation exceeding 0.95 over the sample period.

3.2.5 Volatility Index (VIX_t)

The VIX index is taken as the daily closing level of the CBOE Volatility Index (Yahoo ticker ^VIX). Following Guyot et al. (2026), we use the level (not first-difference) because the ADF test confirms stationarity at conventional significance levels (Brunnermeier et al., 2009). The VIX measures the market's expectation of 30-day S&P 500 volatility and serves as the primary proxy for global equity market fear.

3.2.6 FX Liquidity ($Spread_t$)

The original study (Guyot et al., 2026) employs the daily JPY/USD bid-ask spread from the Bloomberg Terminal. As OANDA API access is pending, we construct a proxy from Yahoo Finance's USDJPY=X daily data: $Spread_t = High_t - Low_t$, expressed in yen. The intraday high-low range serves as a standard proxy for effective bid-ask spread in market microstructure literature (Corwin & Schultz, 2012; Goyenko et al., 2009), capturing intraday liquidity costs when true quote data is unavailable. Should OANDA API access be obtained, we will replace this proxy with true bid-ask quotes for robustness. Formally,

$$Spread_t = High_t^{USDJPY} - Low_t^{USDJPY} \quad (5)$$

expressed in yen. A larger spread indicates lower FX market liquidity and higher transaction costs. Following Bollerslev and Melvin (1994), wider spreads during volatile periods reflect market makers' compensation for heightened adverse selection risk. Gagnon and Chaboud (2007) document that spreads widen precisely during carry trade unwinding episodes, making this proxy essential for our liquidity channel. A larger spread indicates lower FX market liquidity and higher transaction costs. Following Bollerslev and Melvin (1994), wider spreads during volatile periods reflect market makers' compensation for heightened adverse selection risk. Gagnon and Chaboud (2007) document that spreads widen precisely during carry trade unwinding episodes, making this proxy essential for our liquidity channel.

3.2.7 FX Risk Reversal ($RiskReversal_t$)

The original study (Guyot et al., 2026) employs the 1-month JPY/USD 25-delta risk reversal from the Bloomberg Terminal at daily frequency. Publicly available data for this instrument is limited: the BIS Statistics Explorer publishes risk reversal data at quarterly frequency only. We therefore download quarterly JPY/USD 25-delta risk reversal data from the BIS Statistics Explorer web interface and interpolate to daily frequency via cubic spline. This approach follows the methodology described in Guyot et al. (2026, Table 1), which notes that the risk reversal “may need interpolation from quarterly BIS data to daily frequency if intra-month observations are unavailable.” We flag that the interpolated series may underestimate high-frequency variation in hedging costs relative to the original Bloomberg-sourced daily data, and treat this as a data limitation. Formally, the 1-month JPY/USD 25-delta risk reversal is defined as

$$RR_t = \sigma_t^{25\Delta, \text{Call}} - \sigma_t^{25\Delta, \text{Put}} \quad (6)$$

where $\sigma_t^{25\Delta}$ is the implied volatility of a 25-delta out-of-the-money option. A positive (negative) risk reversal signals that call (put) options command a higher premium, indicating market expectations of yen depreciation (appreciation). This can be interpreted as the hedging cost of directional yen exposure (Guyot et al., 2026). Brunnermeier et al. (2009) demonstrate that risk

reversals predict carry trade returns because they capture the skewness (crash risk) embedded in currency option prices.

3.2.8 Equity Performance Gap ($EGAP_t$)

Let SPX_t and NKY_t be the daily closing levels of the S&P 500 and Nikkei 225 stock indices, respectively. The equity performance gap is

$$EGAP_t = 100 \times [\ln(SPX_t) - \ln(SPX_{t-1})] - 100 \times [\ln(NKY_t) - \ln(NKY_{t-1})] \quad (7)$$

representing the daily return differential between US and Japanese equity markets. A positive EGAP means US equities outperform Japanese equities. According to the Uncovered Equity Parity (UEP) hypothesis, this triggers portfolio rebalancing toward yen-denominated assets as investors hedge their increased Dollar exposure, causing yen appreciation (Hau & Rey, 2006; Kim, 2011).

3.2.9 Geopolitical Risk Index (GPR_t)

We download Matteo Iacoviello’s daily Geopolitical Risk (GPR) index from his academic website. The index counts newspaper articles mentioning geopolitical tensions as a share of total articles. We use it exclusively for subsample segmentation rather than as a VAR regressor, classifying each day as “High Stress” ($GPR > \text{historical 90th percentile}$) or “Normal”.

3.3 Descriptive Statistics

Table 2 reproduces the summary statistics reported by Guyot et al. (2026, Table 1) for their sample period of January 1, 2018 through December 31, 2024 ($N = 1,640$ daily observations). These data were assembled from the Bloomberg Terminal. The JPY/USD exchange rate level averaged approximately 123 over their sample, with a daily return near zero (mean 0.02%, SD 0.56%). The interest rate differential remains negative throughout (mean -2.41 percentage points), reflecting Japan’s persistently accommodative monetary stance relative to the US Fed-

eral Reserve. The VIX averages 19.87 but exhibits substantial variation (range 9.22 to 82.69), confirming the sample spans both tranquil and crisis regimes. The 1-month risk reversal mean of -1.05 (range -9.32 to 0.53) indicates persistent demand for yen put protection, consistent with the yen's role as a funding currency for carry trades. The EGAP mean near zero suggests US and Japanese equity markets perform similarly on average over the long run.

Table 2: Summary Statistics Reported by Guyot et al. (2026) (Jan 2018 – Dec 2024, $N = 1,640$)

Variable	N	Mean	SD	Min	Max	ADF p -value
$\Delta JPY/USD$ (%)	1,640	0.02	0.56	-3.86	3.17	<0.01
$\Delta(i - i^*)$ (pp)	1,640	0.00	0.06	-0.75	0.85	<0.01
$\Delta YCurve^{Diff}$ (pp)	1,640	0.00	0.04	-0.51	0.22	<0.01
$\Delta Brent$ (%)	1,640	0.01	1.80	-16.84	8.62	<0.01
VIX (level)	1,640	19.87	7.72	9.22	82.69	<0.01
Spread (yen)	1,640	0.05	0.05	0.00	0.45	<0.01
RiskReversal (pp)	1,640	-1.05	0.82	-9.32	0.53	<0.01
EGAP (%)	1,640	0.00	1.03	-5.23	4.89	<0.01

Note: Reproduced exactly from Guyot et al. (2026, Table 1). All variables are stationary at the 1% level (ADF test). pp = percentage points. Our extended sample (through May 2026) will update these values upon data collection.

All series in the Guyot et al. (2026) sample are stationary at the 1% significance level, satisfying the VAR stability condition. The $\Delta YCurve^{Diff}$ and $\Delta(i - i^*)$ series exhibit means near zero, confirming that first-differencing removed the unit root present in the raw interest rate levels. We will compute equivalent statistics for our extended sample (including the GPR index and the 2025–2026 period) after collecting the data via the public APIs described in Table 1.

3.4 Econometric Framework

We employ a reduced-form Vector Autoregression (VAR) model to capture the dynamic interrelationships among the eight endogenous variables. The VAR methodology is well-suited to this setting because it treats all variables as jointly endogenous, allowing us to trace how a shock to any single variable propagates through the system over time. This approach follows the standard time-series framework for exchange rate analysis established by Guyot et al. (2026) and is

consistent with the broader VAR literature on UIRP deviations (Eichenbaum & Evans, 1995; Engel, 2016).

3.4.1 Baseline VAR Specification

The reduced-form VAR model of lag order p is specified as follows.

$$Y_t = \alpha + \sum_{j=1}^p \Gamma_j Y_{t-j} + e_t \quad (8)$$

where Y_t is the 8×1 vector of endogenous variables defined in equation (9), α is an 8×1 vector of intercept terms, Γ_j is an 8×8 matrix of autoregressive coefficients for lag j , and e_t is an 8×1 vector of white-noise error terms with zero mean and covariance matrix Σ_e .

$$Y_t = [\Delta \text{YCurve}_t^{\text{Diff}}, \Delta(i_t - i_t^*), \Delta \text{Brent}_t, \text{VIX}_t, \text{Spread}_t, \text{RiskReversal}_t, \text{EGAP}_t, \Delta \text{JPY/USD}_t]^T \quad (9)$$

3.4.2 Lag Order Selection

The optimal lag length p is determined by estimating VAR models for lags $p = 1, \dots, 10$ and selecting the specification that minimizes the Schwarz Criterion (SC), also known as the Bayesian Information Criterion (BIC). The SC is preferred over the Akaike Information Criterion (AIC) because it imposes a larger penalty on additional parameters, reducing the risk of overfitting in a system with eight endogenous variables. Guyot et al. (2026) report SC-selected lags of $p = 1$ for the full model and $p = 1$ to 6 for nested sub-models (see their Table 2). We follow their procedure and expect $p = 1$ to be selected for the full specification.

If $p = 1$ is selected, the model simplifies to

$$Y_t = \alpha + \Gamma_1 Y_{t-1} + e_t \quad (10)$$

which preserves degrees of freedom (8 coefficients per equation, plus the intercept) while capturing the first-order autocorrelation structure characteristic of daily financial data.

3.4.3 Stationarity and Stability

Before estimating, we confirm that all variables in Y_t are $I(0)$ via the ADF test (Table 2). After estimation, we verify that all eigenvalues of the companion matrix lie inside the unit circle, a necessary and sufficient condition for VAR stability. If any root lies on or outside the unit circle, we re-estimate with additional differencing or reduce the lag order.

3.4.4 Residual Diagnostics

We test the residuals $\hat{e}_t$ for serial correlation (Breusch-Godfrey LM test), conditional heteroskedasticity (ARCH-LM test), and multivariate normality (Jarque-Bera test). While the normality assumption is routinely violated in daily financial data (leptokurtosis), we proceed with bootstrapped confidence intervals for the impulse response functions to ensure valid inference under non-normal errors, following Guyot et al. (2026).

3.5 Identification Strategy: Cholesky Decomposition

To transform the reduced-form innovations e_t into economically interpretable structural shocks, we employ a recursive identification scheme via the Cholesky decomposition of the residual covariance matrix Σ_e . This approach imposes a Wold causal ordering: a shock to variable k can contemporaneously affect variables $k + 1$ through 8 but not variables 1 through $k - 1$. The ordering is defensible on theoretical grounds rooted in the institutional features of the FX market, the timing of monetary policy decisions, and the microstructure of price discovery across asset classes (Guyot et al., 2026).

Table 3 summarizes the ordering and the implied contemporaneous causality restrictions.

Table 3: Cholesky Decomposition Ordering and Contemporaneous Causality Flow

Position	Variable	Can instantly affect ↓	Is affected instantly by ↑
1	$\Delta YCurve^{Diff}$	Variables 2–8	None
2	$\Delta(i - i^*)$	Variables 3–8	Variable 1
3	$\Delta Brent$	Variables 4–8	Variables 1–2
4	VIX	Variables 5–8	Variables 1–3
5	$Spread$	Variables 6–8	Variables 1–4
6	$RiskReversal$	Variables 7–8	Variables 1–5
7	$EGAP$	Variable 8	Variables 1–6
8	$\Delta JPY/USD$	None	Variables 1–7

Note: Adapted from Guyot et al. (2026). The ordering reflects the theoretical exogeneity hierarchy: monetary policy (positions 1–2) → global risk factors (positions 3–4) → FX market micro-structure (positions 5–6) → equity portfolio rebalancing (position 7) → exchange rate return (position 8).

The economic rationale for each block in the ordering is as follows. The overall structure nests the time-varying risk premium framework within a recursive identification: the risk premium δ_t in Fama’s (1984) decomposition of the forward rate is modeled as a function of the risk, liquidity, and equity indicators occupying positions 3 through 7, consistent with the multi-factor currency risk premium evidence of Nucera et al. (2024).

Monetary Policy Block (Positions 1 and 2): Yield curve differentials ($\Delta YCurve^{Diff}$) and short-term interest rate differentials ($\Delta(i - i^*)$) reflect central bank policy stances. Because the Bank of Japan and Federal Reserve meet at scheduled intervals (approximately 6–8 times per year), daily changes in these variables predate and are orthogonal to same-day FX trading shocks. This exogeneity assumption follows Eichenbaum and Evans (1995), who document that monetary policy responds to exchange rates only with a lag of several months.

Global Risk Block (Positions 3 and 4): Oil price changes ($\Delta Brent$) and equity market volatility (VIX) are determined in global commodity and derivatives markets that operate at a scale far exceeding the JPY/USD market. Japan, as a price-taker in energy markets, cannot influence global oil prices through its exchange rate. The VIX, computed from S&P 500 option prices, reflects aggregate US equity market fear and is plausibly exogenous to JPY-specific liquidity conditions (Brunnermeier et al., 2009).

FX Market Microstructure Block (Positions 5 and 6): The JPY/USD bid-ask spread (*Spread*) represents FX transaction liquidity. Following Engel (2016), liquidity responds to global risk conditions but does not contemporaneously cause them — financial stress widens spreads, but wider spreads do not cause global volatility. The FX risk reversal (*Risk Reversal*) captures the skewness premium in yen options, which represents the marginal cost of hedging yen exposure. Since hedging demand intensifies after liquidity deterioration (and not vice versa), the spread is ordered before the risk reversal (Gagnon & Chaboud, 2007).

Equity Portfolio Rebalancing Block (Position 7): The equity performance gap (*EGAP*) captures the daily return differential between S&P 500 and Nikkei 225. According to the Uncovered Equity Parity (UEP) hypothesis developed by Hau and Rey (2006), a positive EGAP (US outperformance) triggers portfolio rebalancing toward yen-denominated assets as international investors hedge their amplified Dollar exposure. EGAP is endogenous to global risk and liquidity conditions (common factor risks affect both equity markets), but exogenous to the contemporaneous FX return because daily FX order flow — the true driver of exchange rates — is only observable to dealers after the trading session closes (Ding & Ma, 2013; Evans & Lyons, 2002).

Exchange Rate Return (Position 8): The JPY/USD daily return is placed last as the most endogenous variable — the market-clearing price that absorbs all contemporaneous shocks from the preceding seven variables. This is consistent with the FX microstructure view that the exchange rate is the residual price that clears order flow from all customer segments (leveraged investors, unleveraged investors, and non-financial corporations) at the end of each trading day (Evans & Lyons, 2002; Lyons, 2001).

Table 4 provides a consolidated mapping of each variable to its economic concept, theoretical premise, and mathematical formulation.

Table 4: Economic Concepts, Theoretical Premises, and Proxy Formulations

Pos.	Variable	Economic Concept / Theoretical Premise	Formulation
1	$\Delta YCurve_t^{Diff}$	Lagged Monetary Policy Transmission (Eichenbaum & Evans, 1995)	$\Delta(YC^{JPN} - YC^{US})_t$
2	$\Delta(i_t - i_t^*)$	Interest Rate Spread (Carry Trade Potential) (Brunnermeier et al., 2009)	$\Delta(i_t^{JPN} - i_t^{USA})_t$
3	$\Delta Brent_t$	Global Commodity Risk / Terms of Trade (Habib & Stracca, 2012)	$100 \times \ln(Brent_t / Brent_{t-1})$
4	VIX_t	Equity Market Fear / Risk Aversion (Brunnermeier et al., 2009)	S&P 500 30-day implied volatility
5	$Spread_t$	FX Transaction Liquidity / Information Asymmetry (Bollerslev & Melvin, 1994)	$Ask_t - Bid_t$
6	$RiskReversal_t$	JPY Crash Risk / Hedging Cost (Gagnon & Chaboud, 2007)	$\sigma_{25\Delta}^{Call} - \sigma_{25\Delta}^{Put}$
7	$EGAP_t$	Uncovered Equity Parity / Portfolio Rebalancing (Hau & Rey, 2006)	$100 \times [r^{SPX} - r^{NKY}]$
8	$\Delta JPY/USD_t$	FX Market-Clearing / Microstructure (Evans & Lyons, 2002)	$100 \times \ln(s_t / s_{t-1})$

Note: Each variable's position is justified by the cited theoretical framework. $YC^{JPN} = y_{10Y}^{JGB} - y_{2Y}^{JGB}$; $YC^{US} = y_{10Y}^{TNOTE} - y_{2Y}^{TNOTE}$. Returns (r) are daily log-differences in percent.

3.6 Dynamic Analysis Procedures

After estimating the reduced-form VAR and computing the Cholesky factor, we conduct three complementary analyses to characterize the transmission of financial risk shocks to the JPY/USD exchange rate.

3.6.1 Granger Causality Tests

We evaluate predictive relationships by testing the null hypothesis that lagged values of variable x do not contain statistically significant information for forecasting variable y within the VAR system, i.e., H_0 : all coefficients on lags of x in the equation for y are jointly zero. The test statistic follows an F -distribution under the null. We focus on four hypotheses central to our causal claim:

1. Risk indicators ($\Delta Brent$, VIX , $RiskReversal$) do NOT Granger-cause $\Delta JPY/USD$
($H_{GC(1)}$: reject $\Rightarrow$ risk $\rightarrow$ FX)
2. $\Delta(i - i^*)$ does NOT Granger-cause $\Delta JPY/USD$ ($H_{GC(2)}$: accept $\Rightarrow$ UIRP failure)
3. $EGAP$ does NOT Granger-cause $\Delta JPY/USD$ ($H_{GC(3)}$: reject $\Rightarrow$ UEP channel)
4. Risk indicators do NOT Granger-cause $EGAP$ ($H_{GC(4)}$: reject $\Rightarrow$ risk $\rightarrow$ equity $\rightarrow$ FX chain)

These tests follow the identical procedure used by Guyot et al. (2026) (see their Table 3, p. 19). Rejection of $H_{GC(1)}$ and $H_{GC(3)}$ together with acceptance of $H_{GC(2)}$ would confirm that

financial risk channels, rather than interest rate differentials, are the primary predictors of short-run yen movements.

3.6.2 Orthogonalized Impulse Response Functions (IRFs)

To quantify the magnitude, direction, and persistence of risk transmission, we compute orthogonalized impulse response functions. Specifically, we trace the response of $\Delta JPY/USD_{t+h}$ for $h = 0, 1, \dots, 7$ days (one trading week) following a one-standard-deviation positive shock to each of the first seven variables in Y_t , while holding all other contemporaneous shocks fixed at zero via the Cholesky factorization. The orthogonalized IRF for variable j at horizon h is

$$\text{IRF}_{j \rightarrow 8}(h) = \Psi_h P e_j \quad (11)$$

where Ψ_h is the 8×8 matrix of reduced-form moving-average coefficients at horizon h , P is the lower Cholesky factor of Σ_e (i.e., $PP' = \Sigma_e$), and e_j is a selection vector with 1 in the j -th position and 0 elsewhere (L'utkepohl, 2005).

We construct 95% confidence intervals via residual-based bootstrap with $B = 1,000$ replications, following Guyot et al. (2026). For each bootstrap iteration, we: (1) draw T residuals with replacement from $\hat{e}_t$, (2) generate pseudo-data using the estimated VAR coefficients, (3) re-estimate the VAR on the pseudo-data, and (4) recompute the IRFs. The confidence interval is the percentile interval $[\hat{\theta}_{(0.025)}^*, \hat{\theta}_{(0.975)}^*]$ where $\hat{\theta}_{(\alpha)}^*$ is the α -quantile of the bootstrap distribution. An IRF is deemed statistically significant at the 5% level if the 95% confidence band excludes zero (Efron & Tibshirani, 1994).

A negative IRF of $\Delta JPY/USD$ to a VIX shock (yen appreciation when fear rises) would support hypothesis H2.a, confirming the yen's safe-haven role and the carry trade unwinding channel. A negative IRF to an EGAP shock would support hypothesis H4, confirming the UEP portfolio rebalancing channel.

3.6.3 Forecast Error Variance Decomposition (FEVD)

To assess the relative importance of each shock in explaining exchange rate variation, we decompose the h -step-ahead forecast error variance of $\Delta JPY/USD$ into the proportion attributable to each orthogonal shock. The FEVD for variable j at horizon h is

$$FEVD_{j \rightarrow 8}(h) = \frac{\sum_{l=0}^{h-1} (\Psi_l P e_j)^2}{\sum_{l=0}^{h-1} (\Psi_l \Sigma_e \Psi_l')_{88}} \quad (12)$$

which reports the fraction of total forecast error variance of the exchange rate accounted for by innovations to variable j at horizon h (L’utkepohl, 2005). We report FEVD at $h = 7$ days to match the IRF horizon. If the 1-month risk reversal and VIX together account for more than 25% of FEVD (as found by Guyot et al. (2026), Figure 3), this would confirm the dominance of risk sentiment over interest rates in driving short-term exchange rate movements.

3.7 Robustness Checks

3.7.1 Alternative Cholesky Orderings

Because daily financial variables exhibit high contemporaneous correlation, the Cholesky decomposition results may be sensitive to the assumed variable ordering. To test the robustness of our findings, we estimate five alternative specifications that permute the ordering of the risk and equity variables, following Guyot et al. (2026, pp. 29–30):

1. **Risk-first ordering:** Global risk variables ($\Delta Brent$, VIX) precede monetary policy variables, testing the sensitivity of the exogenous block.
2. **EGAP before risk:** The equity performance gap is placed before market risk indicators, testing whether equity returns drive risk rather than the reverse.
3. **Risk reversal before spread:** FX market risk precedes FX liquidity, testing whether hedging demand determines liquidity conditions.

4. **Monetary policy endogenous:** Yield curve and interest rate differentials are placed after all financial variables, relaxing the strong exogeneity assumption.
5. **Exchange rate as intermediate:** The exchange rate return is placed in the middle of the ordering, testing whether FX returns contemporaneously affect risk variables.

If the signs and statistical significance of the VIX and risk reversal IRFs remain stable across these specifications, we can be confident that our conclusions are not artifacts of a particular ordering.

3.7.2 Subsample Stability Analysis

Financial risk transmission channels may exhibit nonlinear behavior across different market regimes. Following Guyot et al. (2026), we re-estimate the baseline VAR on four subsamples:

1. **High-stress days:** Observations in the top quartile of VIX (threshold ≈ 22.86).
2. **Low-stress days:** Observations in the remaining three quartiles.
3. **Pre-2022 period:** January 2018 – December 2021 (pre-tightening regime).
4. **Post-2022 period:** January 2022 – May 2026 (Fed tightening / Iran crisis).

In addition, we introduce a **Geopolitical Risk (GPR) subsample** that partitions the data using the GPR index's 90th percentile threshold. This directly tests whether the Iran conflict period amplifies the carry trade unwind channel relative to normal times. Habib and Stracca (2012) and Lee (2017) predict that safe-haven effects should be strongest during moderate crises but may attenuate during tail-risk events; comparing IRFs across GPR regimes allows us to assess this nonlinearity directly.

3.8 Methodological Adjustments Relative to the Source Study

While our methodology closely follows Guyot et al. (2026), we introduce three modifications to adapt the framework to our research question and data environment.

Extension 1 – Extended Temporal Scope. The original sample ends December 2024. We extend through May 2026 to capture the Iran conflict period. This extension adds approximately 400 trading days that include extreme observations for VIX, Brent, GPR, and the JPY/USD exchange rate, providing a natural stress-test of the baseline model’s stability.

Extension 2 – GPR Regime Segmentation. The original study uses VIX-quartile and pre-/post-2022 subsamples. We add a GPR-based segmentation that provides a geopolitical (rather than purely financial) classification of stress periods. This isolates the Iran war’s effect from general financial volatility.

Modification 3 – Public Data Substitution. The original study accesses all data via Bloomberg Terminal. Our replication uses only freely accessible public sources: FRED (via `pandas_datareader`), Yahoo Finance (via `yfinance`), MOF Japan (daily JGB yield fixings), and the BIS Statistics Explorer. Three specific substitutions merit explicit acknowledgement. First, the Japan Call Rate used by Guyot is not available daily from any public API; we substitute the daily 1-Year JGB yield from MOF Japan as a short-term rate proxy, and use the Fed Funds Rate (DFF, daily from FRED) for the US rather than the 3-month T-bill, matching the instrument Guyot actually employed. Second, the JPY/USD bid-ask spread is substituted with the intraday range proxy (High – Low) from Yahoo Finance’s USDJPY=X data, a standard liquidity proxy in the market microstructure literature (Corwin & Schultz, 2012; Goyenko et al., 2009), pending OANDA API access. Third, the 1-month JPY/USD 25-delta risk reversal is published by the BIS only at quarterly frequency; we interpolate to daily via cubic spline and flag this as a data limitation relative to the original study’s Bloomberg-sourced daily data. These substitutions are described in detail in Table 1 and the accompanying Data Substitution Note.

3.9 Ideal Experiment vs. Empirical Research Design

An ideal experiment to identify the causal effect of financial risk on the yen would randomly assign a geopolitical crisis while holding constant all other global and domestic macroeconomic conditions. In this ideal laboratory, researchers would exogenously shock the VIX and risk

reversals, then observe the exchange rate response, directly measuring the two mediation channels through proprietary broker order-flow data that captures carry trade liquidation volumes and equity hedge rebalancing flows.

Since such an experiment is infeasible, our research design approximates it through three features. First, the March 2026 Iran conflict serves as a plausibly exogenous shock originating outside Japan’s economic fundamentals — it originates from geopolitical tensions in the Middle East, not from Japanese monetary or fiscal policy. Second, the recursive (Cholesky) identification isolates the component of each shock that is orthogonal to preceding variables, approximating the ideal “randomized” treatment. Third, the GPR subsample analysis compares the VAR impulse responses across crisis and non-crisis periods, mimicking a difference-in-differences logic where the 2025–2026 period is the treatment group and 2018–2024 is the control group. This quasi-experimental triangulation strengthens the credibility of our causal claims, even though we cannot fully rule out omitted variable bias from unobserved simultaneous shocks (Angrist & Pischke, 2009).

3.10 Software Implementation and Reproducibility

All analyses are conducted in **R** (version ≥ 4.3) and **Python** (version ≥ 3.10), using the following key packages.

Data Retrieval: `pandas_datareader` (Python) for FRED data via the CSV export endpoint; `yfinance` (Python) for Yahoo Finance data; MOF Japan daily JGB yield fixings retrieved via direct CSV download from the Ministry of Finance website; Iacoviello GPR index downloaded as XLSX and converted to CSV; BIS risk reversal data retrieved manually from the BIS Statistics Explorer web interface; `quantmod` (R) for supplementary downloads and data alignment.

Time Series Analysis: `vars` (R) for VAR estimation, lag selection, Granger causality, IRF, and FEVD; `urca` (R) for ADF unit root tests; `tseries` (R) for ARCH-LM and Jarque-Bera diagnostics; `lmtest` (R) for Breusch-Godfrey serial correlation test.

Sandwich Inference: `sandwich` (R) for heteroskedasticity- and autocorrelation-consistent standard errors where needed.

To maximize reproducibility, we will release a complete GitHub repository containing:

- Python scripts for data retrieval from FRED and Yahoo Finance APIs
- R scripts for data alignment, variable construction, VAR estimation, IRF/FEVD computation, and robustness checks
- The final aligned dataset (CSV) and a Jupyter notebook demonstrating the full pipeline end-to-end
- A `README.md` with installation instructions and a `requirements.txt` / `renv.lock` for exact package versioning

All code, data, and documentation will be archived at a persistent URL (e.g., Zenodo or OSF) upon completion. This commitment to open science follows best practices in financial econometrics and ensures that any researcher can independently verify, replicate, and extend our findings (Rossi, 2013).

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